

Joint AoA and AoD Estimation for ISAC in mmWave Hybrid Arrays with Covariance Matrix Reconstruction and Practical Validation

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Abstract—With the evolution toward sixth-generation (6G) mobile communications, millimeter-wave (mmWave) hybrid analog-digital arrays have become pivotal in integrated sensing and communication (ISAC) systems. However, the inherent dimensionality reduction from the element space to beam space significantly complicates high-precision angle estimation. To address this problem, we first introduce an orthogonal frequency division multiplexing (OFDM)-based delay-domain filtering mechanism to suppress dense clutter and interference in rich-scattering environments. Building upon the filtered observations, we propose a joint angle of arrival (AoA) and angle of departure (AoD) estimation algorithm based on bidirectional covariance matrix reconstruction (Bi-CMR). The proposed method effectively recovers the complete array manifold from compressed observations by leveraging the linear mapping between beam space channel power and the vectorized full-dimensional channel covariance matrix. Simulations show the algorithm outperforms conventional methods under the low signal-to-noise ratio (SNR) and multipath conditions. Experimental results confirm high-accuracy angle estimation in both monostatic and distributed deployments.

Index Terms—ISAC, Hybrid Analog-Digital Arrays, Joint AoA and AoD Estimation, Covariance Matrix Reconstruction.

I. INTRODUCTION

The shift toward high-frequency communication necessitates the use of massive multiple-input multiple-output (MIMO) to counteract severe path loss and absorption. In millimeter-wave (mmWave) bands, the hybrid analog-digital beamforming (HAD-BF) architecture has gained dominance as a cost-effective alternative to fully digital array systems [1]. As the sixth-generation (6G) mobile communications evolves, integrated sensing and communication (ISAC) has emerged as a new paradigm, requiring the dual-use of hardware and spectral resources for both data transmission and environmental sensing. A key challenge in ISAC is achieving high-precision

angle estimation under the constraints of HAD-BF architectures. Since precise angular information is vital for both target localization and beam management, researching robust angle estimation algorithms for hybrid arrays is essential for the realization of high-performance ISAC systems [2].

Despite their potential, HAD-BF systems present unique challenges for angle estimation. Unlike traditional digital arrays where the extended multiple signal classification (MUSIC) and estimating signal parameters via rotational invariance techniques (ESPRIT) algorithms can be directly applied to the full-dimensional signals, HAD-BF architectures suffer from a significant loss in spatial degrees of freedom (DoF) due to the reduced number of radio frequency (RF) chains. To address this dimensionality reduction, existing research has explored various paradigms, including beamforming-based, subspace-based, sparsity-aware, and data-driven algorithms. Analog phase alignment (APA) and exhaustive traversal methods [3], [4], estimate angles by observing the received signal strength across different beam directions. Although computationally efficient, these methods are fundamentally limited by the Rayleigh criterion and the codebook resolution. The monopulse angle estimation algorithm [5] is a typical algorithm but relies on prior information about the target's direction. To achieve super-resolution, subspace-based methods like extended MUSIC and covariance matrix reconstruction (CMR) have been proposed. The extended MUSIC algorithm called H-MUSIC [6] directly utilizes the received echo signal to distinguish between the signal subspace and the noise subspace. Although simple, H-MUSIC loses the original covariance structure information. CMR-based approaches [7] can restore the original covariance structure from limited measurements, offering a robust balance between complexity and estimation fidelity. They can be combined with MUSIC or ESPRIT to generate high-precision angle estimates [8]. Recent advancements have further exploited the Toeplitz structure to reduce reconstruction complexity from $\mathcal{O}(N^3)$ to $\mathcal{O}(N \log N)$ [9]. Compressed sensing (CS) algorithms [10] leverage the sparsity of massive array echoes in the angular domain, transforming the angle estimation problem into a sparse reconstruction problem solved using tools such as compressed sensing or convex optimization. Neural network-based estimation algorithms [11] represent a novel approach, transforming traditional parameter estimation problems into data-driven learning processes, achieving good estimation performance, though issues such as model generalization need to be considered.

However, several critical gaps remain in the existing litera-

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ture. First, most current research focuses solely on the angle of arrival (AoA) at the receiver, whereas ISAC applications increasingly demand joint estimation of both the AoA and angle of departure (AoD) to fully characterize target scattering. Second, although mmWave channels are physically sparse, the number of environmental scatterers in complex scenarios can easily exceed the limited spatial DoFs of a hybrid array, leading to a saturated subspace and severe performance degradation.

To bridge these gaps, this letter proposes a joint AoA and AoD bidirectional angle estimation framework for mmWave HAD-BF architectures. Specifically, we develop an orthogonal frequency division multiplexing (OFDM)-based delay-domain filtering mechanism that exploits multipath sparsity to effectively suppress dense clutter, thereby providing purified signal observations for subsequent processing. Subsequently, a bidirectional covariance matrix reconstruction (Bi-CMR) algorithm is derived to restore the full-dimensional equivalent covariance matrix from compressed beam space data, which successfully bypasses the spatial degree-of-freedom (DoF) limitations imposed by sparse RF chains. Furthermore, the effectiveness and robustness of the proposed framework are validated through extensive experimental evaluations using 25.05 GHz mmWave active antenna units (AAUs) in complex scattering environments.

II. SYSTEM MODEL

Consider a MIMO-OFDM communication system where the transceivers operate under a HAD-BF architecture. Without loss of generality, we consider a single RF-chain configuration, where the transmitter and receiver utilize uniform linear arrays (ULAs) with M_t and M_r antenna elements, respectively. The system employs K subcarriers in total. For the k -th subcarrier, when the transmitter employs the transmit beamforming vector $\mathbf{f}_q \in \mathbb{C}^{M_t \times 1}$ and the receiver employs the receive beamforming vector $\mathbf{w}_p \in \mathbb{C}^{M_r \times 1}$, the frequency-domain received signal can be expressed as

$$y_{p,q}[k] = \mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q s[k] + \mathbf{w}_p^T \mathbf{z}[k], \quad (1)$$

where $s[k]$, $\mathbf{H}[k]$, and $\mathbf{z}[k]$ denote the transmitted symbol, frequency-domain channel matrix, and additive white Gaussian noise on subcarrier k , with $\mathbb{E}[\mathbf{z}[k]\mathbf{z}^H[k]] = N_0 \mathbf{I}_{M_r}$. The beamforming vectors $\mathbf{f}_q$ and $\mathbf{w}_p$ are selected from predefined codebooks $\mathcal{F} = \{\mathbf{f}_1, \mathbf{f}_2, \dots, \mathbf{f}_{Q_t}\}$ and $\mathcal{W} = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{Q_r}\}$, with Q_t and Q_r representing the total number of transmit and receive scanning directions.

In view of the sparse scattering nature of mmWave propagation, we adopt a geometric physical model for the frequency-domain channel. For the k -th subcarrier, the channel matrix $\mathbf{H}[k]$ is modeled as a sum of L discrete paths:

$$\mathbf{H}[k] = \sum_{l=0}^{L-1} \alpha_l e^{-j2\pi k \Delta f \tau_l} \mathbf{a}_r(\phi_l) \mathbf{a}_t^T(\theta_l), \quad (2)$$

where α_l , τ_l , ϕ_l , and θ_l denote the complex gain, propagation delay, angle of arrival (AoA), and angle of departure (AoD) of the l -th path component, respectively. $\mathbf{a}_r(\phi_l)$ and $\mathbf{a}_t(\theta_l)$ denote

the corresponding receive and transmit steering vectors. Δf denotes the subcarrier spacing.

Specifically, considering a ULA configuration, receive and transmit steering vectors $\mathbf{a}_r(\phi_l)$ and $\mathbf{a}_t(\theta_l)$ are defined as

$$\mathbf{a}_r(\phi_l) = \left[1, e^{-j2\pi \frac{d \sin(\phi_l)}{\lambda}}, \dots, e^{-j2\pi (M_r-1) \frac{d \sin(\phi_l)}{\lambda}} \right]^T, \quad (3)$$

$$\mathbf{a}_t(\theta_l) = \left[1, e^{-j2\pi \frac{d \sin(\theta_l)}{\lambda}}, \dots, e^{-j2\pi (M_t-1) \frac{d \sin(\theta_l)}{\lambda}} \right]^T. \quad (4)$$

where d denotes the inter-element spacing and λ represents the wavelength of the carrier frequency.

Therefore, by projecting the high-dimensional MIMO channel onto the beam space defined by $\mathbf{f}_q$ and $\mathbf{w}_p$, the resulting low-dimensional channel estimation for the (p, q) -th beam pair is given by

$$c_{p,q}[k] = \mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q + \mathbf{w}_p^T \mathbf{z}[k]. \quad (5)$$

where $\mathbf{w}_p^T \mathbf{z}[k]$ is the effective noise. In practice, $c_{p,q}[k]$ serves as the observation for subsequent angle estimation.

III. PROPOSED SCHEME

A. Multipath Suppression

The extended cancellation algorithm (ECA) [12] is a subspace-based technique extensively utilized in radar systems for clutter suppression. In this work, we leverage the ECA for delay-domain filtering to mitigate unwanted multipath components. It ensures that the number of effective targets remains below the number of antenna elements, which is a prerequisite for high-resolution angle estimation. First, we define the target delay subspace. Given a specific delay range of interest $\mathcal{T}_{\text{target}} = \{\tau_1, \tau_1 + \Delta\tau, \dots, \tau_2\}$, the basis matrix $\mathbf{S} \in \mathbb{C}^{K \times M}$ is constructed using the delay steering vectors as

$$\mathbf{S} = [\mathbf{v}(\tau_1), \mathbf{v}(\tau_1 + \Delta\tau), \dots, \mathbf{v}(\tau_2)], \quad (6)$$

where $\mathbf{v}(\tau_m) = [1, e^{-j2\pi \Delta f \tau_m}, \dots, e^{-j2\pi (K-1) \Delta f \tau_m}]^T$ and $\Delta\tau \leq 1/(K\Delta f)$ denotes the delay sampling interval.

The filtering operation is performed by projecting the received beam space observations onto the subspace defined by $\mathbf{S}$. The corresponding projection matrix is $\mathbf{P}_{\text{target}} = \mathbf{S}(\mathbf{S}^H \mathbf{S})^{-1} \mathbf{S}^H$. Accordingly, the preprocessed signal within $[\tau_1, \tau_2]$ is given by

$$\tilde{\mathbf{c}}_{p,q} = \mathbf{P}_{\text{target}} \mathbf{c}_{p,q}, \quad (7)$$

where $\mathbf{c}_{p,q}$ and $\tilde{\mathbf{c}}_{p,q}$ are $K \times 1$ vectors collecting the frequency-domain samples $c_{p,q}[k]$ and $\tilde{c}_{p,q}[k]$ for $k = 1, \dots, K$, respectively.

B. Proposed Bi-CMR algorithm and angle estimation

To focus on extracting angular information and mitigate the impact of small-scale fading and noise, we utilize frequency-domain smoothing across all K subcarriers. Assuming $\mathbf{z}[k]$ and the channel $\mathbf{H}[k]$ are independent, we can compute the expected power of the beam space observation $P_{p,q}$ as

$$\begin{aligned} P_{p,q} &= \mathbb{E} [|\tilde{c}_{p,q}[k]|^2] \\ &= \mathbb{E} [(\mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q + \mathbf{w}_p^T \mathbf{z}[k])(\mathbf{f}_q^H \mathbf{H}^H[k] \mathbf{w}_p^* + \mathbf{z}^H[k] \mathbf{w}_p^*)] \\ &= \mathbb{E} [|\mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q|^2] + N_0 \mathbf{w}_p^H \mathbf{w}_p. \end{aligned} \quad (8)$$

where $P_{p,q} = \frac{1}{K} \sum_{k=1}^K |\tilde{c}_{p,q}[k]|^2$ represents the average power across different subcarriers.

Assuming the noise power can be estimated and subtracted from the measurements, we obtain

$$\tilde{P}_{p,q} = P_{p,q} - N_0 \mathbf{w}_p^H \mathbf{w}_p = \mathbb{E} [|\mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q|^2], \quad (9)$$

To facilitate the angle estimation, we first rewrite the scalar term $\mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q$ by employing the vectorization operator. By invoking the Kronecker product identity $\text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\mathbf{B}^T \otimes \mathbf{A})\text{vec}(\mathbf{X})$, the term $\mathbf{H}[k] \mathbf{f}_q$ can be expanded as

$$\text{vec}(\mathbf{H}[k] \mathbf{f}_q) = (\mathbf{f}_q^T \otimes \mathbf{I}_{M_r}) \text{vec}(\mathbf{H}[k]). \quad (10)$$

Consequently, the received signal for the (p, q) -th beam pair can be rewritten in a compact vectorized form:

$$\mathbf{w}_p^T \mathbf{H}[k] \mathbf{f}_q = (\mathbf{f}_q^T \otimes \mathbf{w}_p^T) \text{vec}(\mathbf{H}[k]). \quad (11)$$

Substituting the vectorized signal model into (8), the noise-subtracted power $\tilde{P}_{p,q}$ for the (p, q) -th beam pair is given by:

$$\tilde{P}_{p,q} = (\mathbf{f}_q^T \otimes \mathbf{w}_p^T) \mathbf{R}_H (\mathbf{f}_q^T \otimes \mathbf{w}_p^T)^H, \quad (12)$$

where $\mathbf{R}_H \triangleq \mathbb{E} [\text{vec}(\mathbf{H}[k]) \text{vec}(\mathbf{H}[k])^H]$ denotes the covariance matrix of the vectorized channel. Since $\tilde{P}_{p,q}$ represents the denoised power, $\mathbf{R}_H$ reflects the spatial structure of the propagation environment. Under the assumption of uncorrelated scattering paths, $\mathbf{R}_H$ can be further expanded as

$$\mathbf{R}_H = \sum_{l=0}^{L-1} \sigma_l^2 \mathbf{a}(\phi_l, \theta_l) \mathbf{a}^H(\phi_l, \theta_l), \quad (13)$$

where $\sigma_l^2 = \mathbb{E}[|\alpha_l|^2]$ is the average power of the l -th path, and $\mathbf{a}(\phi_l, \theta_l) \triangleq \mathbf{a}_t(\theta_l) \otimes \mathbf{a}_r(\phi_l) \in \mathbb{C}^{M_t M_r \times 1}$ represents the joint transmit-receive steering vector. Notably, $\mathbf{R}_H$ is a low-rank matrix with $\text{rank}(\mathbf{R}_H) = L$, where L denotes the number of resolved scattering paths.

To cast the power measurement into a linear framework for covariance reconstruction, we need to convert the quadratic form into a linear form. Using the properties of the trace and the vectorization identity $\text{tr}(\mathbf{A}^H \mathbf{B}) = \text{vec}(\mathbf{A})^H \text{vec}(\mathbf{B})$, we have:

$$\begin{aligned} \tilde{P}_{p,q} &= \text{tr}((\mathbf{f}_q^T \otimes \mathbf{w}_p^T) \mathbf{R}_H (\mathbf{f}_q^* \otimes \mathbf{w}_p^*)) \\ &= \text{tr}((\mathbf{f}_q^* \otimes \mathbf{w}_p^*) (\mathbf{f}_q^T \otimes \mathbf{w}_p^T) \mathbf{R}_H) \\ &= \text{vec}((\mathbf{f}_q^* \otimes \mathbf{w}_p^*) (\mathbf{f}_q^T \otimes \mathbf{w}_p^T))^H \text{vec}(\mathbf{R}_H). \end{aligned} \quad (14)$$

Furthermore, by applying the property $\text{vec}(\mathbf{a}\mathbf{b}^H) = \mathbf{b}^* \otimes \mathbf{a}$, the vectorized term in (14) can be expanded as

$$\tilde{P}_{p,q} = [(\mathbf{f}_q^H \otimes \mathbf{w}_p^H) \otimes (\mathbf{f}_q^T \otimes \mathbf{w}_p^T)] \text{vec}(\mathbf{R}_H). \quad (15)$$

Through the above derivation, we have proven that for each transmit-receive beam pair $\mathbf{f}_q$ and $\mathbf{w}_p$, there exists a deterministic linear relationship between the measured channel contribution power $\tilde{P}_{p,q}$ and the vectorized form of the full channel covariance matrix $\mathbf{R}_H$:

$$\tilde{P}_{p,q} = \mathbf{a}_{p,q} \mathbf{r}_H, \quad (16)$$

where $\mathbf{r}_H = \text{vec}(\mathbf{R}_H)$ and $\mathbf{a}_{p,q} = (\mathbf{f}_q^H \otimes \mathbf{w}_p^H) \otimes (\mathbf{f}_q^T \otimes \mathbf{w}_p^T)$.

By aggregating all $Q_t Q_r$ beam-pair measurements, the core linear system of the proposed framework is formulated as

$$\mathbf{A} \mathbf{r}_H = \tilde{\mathbf{p}}, \quad (17)$$

where each row of $\mathbf{A}$ corresponds to $\mathbf{a}_{p,q}$, and each element of $\tilde{\mathbf{p}}$ corresponds to $\tilde{P}_{p,q}$.

To ensure numerical stability when $\mathbf{A}$ is ill-conditioned, the vectorized covariance matrix is estimated via the regularized least-squares (RLS) criterion, also known as diagonal loading

$$\hat{\mathbf{r}}_H = (\mathbf{A}^H \mathbf{A} + \sigma^2 \mathbf{I}_{M_t M_r})^{-1} \mathbf{A}^H \tilde{\mathbf{p}}, \quad (18)$$

where σ^2 denotes the regularization (diagonal loading) factor. To satisfy the fundamental property of a physical covariance matrix, we enforce the Hermitian symmetry as

$$\hat{\mathbf{R}} = \frac{1}{2} (\text{unvec}(\hat{\mathbf{r}}_H) + \text{unvec}(\hat{\mathbf{r}}_H)^H). \quad (19)$$

Finally, the AoA and AoD are extracted through subspace-based spectral estimation. Performing eigenvalue decomposition on $\hat{\mathbf{R}}$ yields:

$$\hat{\mathbf{R}} = \mathbf{U}_s \mathbf{\Lambda}_s \mathbf{U}_s^H + \mathbf{U}_n \mathbf{\Lambda}_n \mathbf{U}_n^H \quad (20)$$

where $\mathbf{U}_s \in \mathbb{C}^{M_t M_r \times L}$ represents the signal subspace, $\mathbf{U}_n \in \mathbb{C}^{M_t M_r \times (M_t M_r - L)}$ represents the noise subspace, $\mathbf{\Lambda}_s = \text{diag}(\lambda_1, \dots, \lambda_L)$ with λ_i denoting the i -th dominant eigenvalue of $\hat{\mathbf{R}}$. In contrast, $\mathbf{\Lambda}_n = \text{diag}(\lambda_{L+1}, \dots, \lambda_{M_t M_r})$ consists of the remaining minor eigenvalues, which correspond to the noise floor and are ideally close to zero in high signal-to-noise ratio (SNR) conditions.

Thus, the angles of departure and arrival can be estimated via spectral peak search

$$P(\phi, \theta) = |\mathbf{U}_n^H \mathbf{a}(\phi, \theta)|_2^2, \quad \phi, \theta \in [-90^\circ, 90^\circ]. \quad (21)$$

IV. PERFORMANCE SIMULATION

In this section, we evaluate the performance of the proposed angle estimation algorithm through numerical simulations. The system parameters, chosen in accordance with 5G new radio (NR) standard, are detailed in Table I. The spatial scenarios involve multipath components with angles and delays following uniform distributions.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier Frequency	25.05 GHz
Subcarrier Spacing	120 kHz
Number of Effective Subcarriers	1584
System Bandwidth	200 MHz
Number of Transmit Antennas	4
Number of Receive Antennas	4
Number of Transmit Codebooks	7
Number of Receive Codebooks	7
Target Angle Distribution	$\mathcal{U}(-60^\circ, 60^\circ)$
Target Delay Distribution	$\mathcal{U}(0, 226)$ m

The root mean square error (RMSE) performance of the investigated algorithms is compared in Fig. 1. The proposed framework demonstrates a significant 10 dB SNR gain over existing methods when targeting a 1° estimation precision.

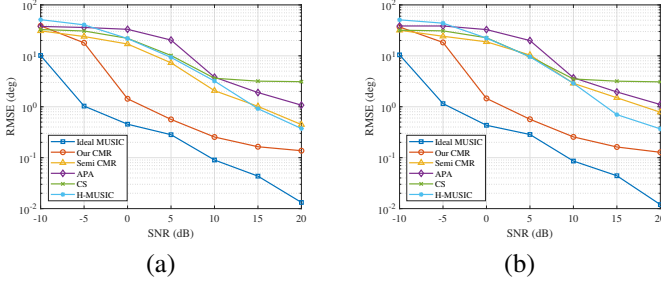


Fig. 1. Estimation error versus SNR. (a) AoA and (b) AoD.

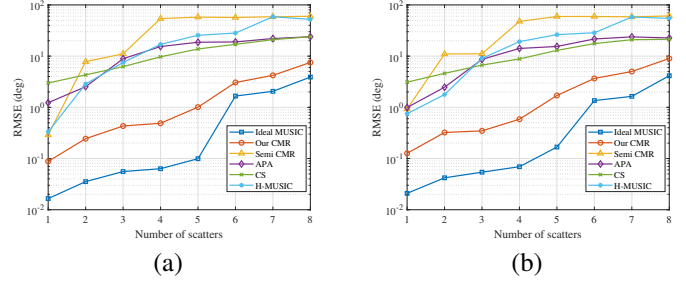


Fig. 2. Estimation error versus the number of targets. (a) AoA and (b) AoD.

By reconstructing the full-dimensional covariance matrix, our method maximizes the transmit-receive spatial diversity, thereby attaining near-optimal performance benchmarked against the full-digital MUSIC algorithm. In contrast, while CS algorithms [10] also utilize joint angular information, their precision is fundamentally capped at high SNR regimes due to grid mismatch and the restricted properties of the sensing matrix. Traditional APA schemes [4], on the other hand, suffer from significant latency due to exhaustive beam sweeping, with their resolution strictly limited by the codebook size. The conventional CMR algorithm [7] called semi-CMR is bottlenecked by the beamforming gain of the fixed-side vector during its sequential estimation process, leading to suboptimal fidelity. Lastly, although the H-MUSIC algorithm [6] offers lower computational complexity, its estimation accuracy and the number of resolvable targets are constrained by the sampling diversity of the beam space signals.

Furthermore, the sensitivity of the estimation accuracy to the number of scatterers is evaluated in Fig. 2. While a general degradation in precision is observed as the number of multipath components increases, the proposed MUSIC-based framework maintains its super-resolution capability and consistently outperforms existing benchmarks. Notably, we observe a sharp performance degradation for the semi-CMR and H-MUSIC algorithms from Fig. 2a when the number of scatterers exceeds 4 and 7, respectively. Similar observations can be made for AoD estimation in Fig. 2b. This phenomenon is attributed to the limited degrees of freedom (DoF) of these methods: the semi-CMR algorithm only reconstructs the $M_r \times M_r$ receive covariance matrix, which can identify at most $M_r = 4$ targets depending on the coherence. Similarly, the H-MUSIC algorithm is constrained by the number of receive beam patterns ($Q_r = 7$). When the number of targets exceeds these DoF limits, the signal and noise subspaces become indistinguishable, leading to estimation failure. In contrast, by exploiting the joint transmit-receive spatial structure, the proposed algorithm yields a virtual equivalent array of size $M_t M_r = 16$, thereby supporting the simultaneous estimation of up to 16 targets. Furthermore, by incorporating delay-domain segmentation filtering, the system's capacity can be further extended to resolve even more targets across different delay bins.

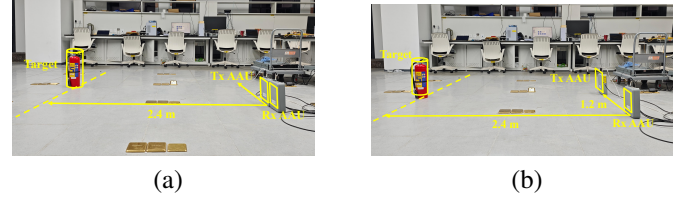


Fig. 3. Millimeter-wave experimental system scenario setup. (a) monostatic deployment and (b) distributed deployment.

V. EXPERIMENTAL EVALUATION

We tested our algorithm on data collected from baseband unit (BBU) using general-purpose hardware platforms and commercial AAUs. The system frequency was set to 25.05 GHz, and the frame structure compliant with the 5G NR standard with a subcarrier spacing of 120 kHz and a slot duration of 0.125 ms. Both the transmit and receive AAUs performed beam traversal using the same codebook, with beam directions set to $[-48^\circ, -25^\circ, -8^\circ, 0^\circ, 8^\circ, 25^\circ, 48^\circ]$ respectively. The transmit AAU and the receive AAU were connected and synchronized via a 1588 switch. The system scenarios are shown in Fig. 3. We tested the performance when the transmitter and receiver were co-located and when they were separated. As shown in Fig. 3a, when the transmitter and receiver are co-located, this represents a typical monostatic radar mode simulated by a communication system. As shown in Fig. 3b, when the transmitter and receiver are separated, this represents a typical multi-station cooperative system or a transmitter-receiver pair in a cell-free mmWave system.

First, the estimation accuracy was evaluated in the monostatic radar mode. As shown in Fig. 3a, we set up 9 test points at intervals of 10° and placed scatterers at each point to measure the angle estimation error. Fig. 4a demonstrates that high-precision estimation is consistently achieved across all test positions. As summarized in Fig. 4b, the mean AoA and AoD estimation errors are 2.6° and 2.1° , respectively. By leveraging the property that AoA equals AoD in this configuration, the direction estimation error can be reduced to 1.3° .

Subsequently, the performance is assessed in the distributed communication mode at the same 9 test points shown in Fig. 3b. As shown in Fig. 5a, high-precision angle estimation can be achieved at all test points, but the AoD estimation error is larger. This phenomenon is primarily attributed to the fact that edge-case targets are situated at large off-boresight angles relative to the transmitter, where beamforming gain

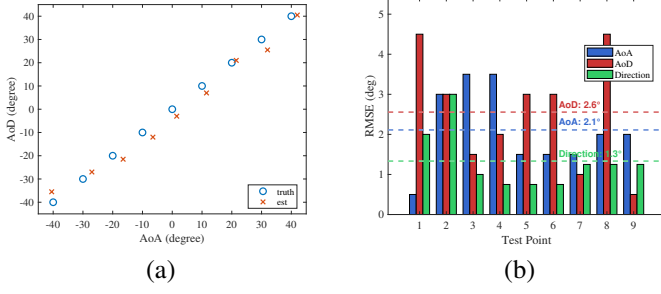


Fig. 4. Angle estimation error at different test points in monostatic deployment. (a) Topology of test points and corresponding angle estimation. (b) Bar chart and mean of the errors.

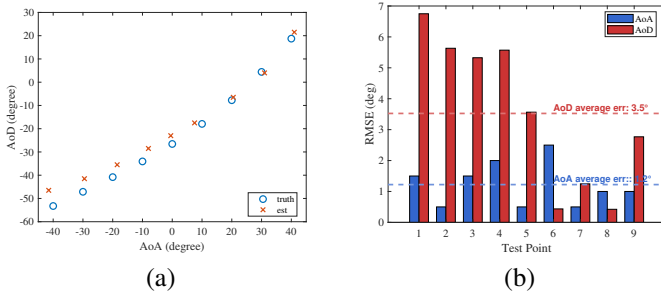


Fig. 5. Angle estimation error at different test points in distributed deployment. (a) Topology of test points and corresponding angle estimation. (b) Bar chart and mean of the errors.

degradation and beam squint effects typically lead to larger pointing errors. Fig. 5b indicates that under this configuration, the mean AoA and AoD estimation errors are recorded as 1.2° and 3.5° , respectively.

To verify the algorithm's effectiveness in multi-target scenarios, we finally assessed its spatial localization performance for multiple scatterers. As shown in Fig. 6, the proposed framework achieves high-precision estimation for both two-target and three-target configurations. Specifically, the observed AoA/AoD errors are $2.0^\circ/1.5^\circ$ for two targets shown in Fig. 6a and $4.7^\circ/2.7^\circ$ for three targets shown in Fig. 6b. The successful recovery of these angular parameters demonstrates the superior multi-target angle estimation capability of our proposed covariance reconstruction approach.

VI. CONCLUSION

This letter investigated the joint estimation of AoA and AoD for mmWave ISAC systems utilizing HAD-BF architectures. To mitigate the interference from dense multipath components, a delay-domain filtering mechanism was proposed, effectively isolating target echoes for improved multi-target discrimination. Furthermore, a BI-CMR algorithm was developed by exploiting the linear mapping between beam space power measurements and the full-dimensional array manifold. By restoring the equivalent spatial covariance from compressed observations, the proposed algorithm overcomes the physical constraints on spatial DoF imposed by the limited number of RF chains. Simulation results demonstrate that high-precision multi-target sensing is achievable with constrained beam-scanning overheads. Experimental evaluations further validate

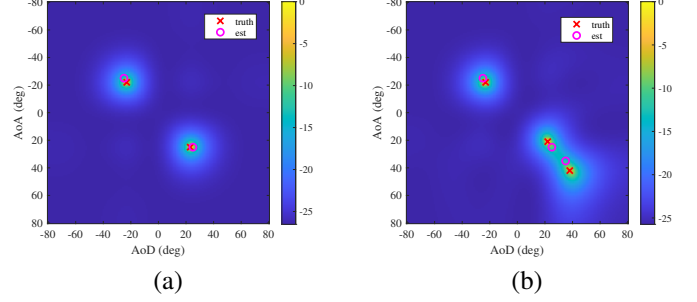


Fig. 6. Angle estimation error in multi-target scenarios. (a) Two targets and (b) Three targets.

the proposed approach, yielding angular estimation accuracies of 1.3° for co-located deployments and 1.5° for distributed deployments, with multi-target estimation errors remaining within 5° . Future research will extend this algorithm to planar arrays and high-mobility scenarios to enhance robustness in low-SNR environments.

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